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—— 梅宏 中国科学院院士/发展中国家科学院院士

以低年级开设“深入理解计算机系统”课程为基础，我先后在复旦大学和上海交通大学软件学院主导了激进的教学改革……现在我课题组的青年教师全部是首批经历此教学改革的学生。本科的扎实基础为他们从事系统软件的研究打下了良好的基础……师资力量的补充又为推进更加激进的教学改革创造了条件。

—— 臧斌宇 上海交通大学软件学院院长

本书是一本将计算机软件和硬件理论结合讲述的经典教程，内容覆盖计算机导论、体系结构和处理器设计等多门课程。本书的最大优点是从程序员的角度描述计算机系统的实现细节，通过描述程序是如何映射到系统上，以及程序是如何执行的，使读者更好地理解程序的行为，以及程序效率低下的原因。

和第2版相比，本版内容上最大的变化是，从以IA32和x86-64为基础转变为完全以x86-64为基础。主要更新如下：

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- 增加了对信号处理程序更细致的描述，包括异步信号安全的函数等。
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